Brain Tumor Detection

Neural Networks for Computer Vision

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Contents / Agenda

1. Requirements Gathering
2. Data Overview
3. Exploratory Data Analysis
4. Data Preprocessing
5. Model Building - Simple Artificial Neural Networks
6. Model Building - Optimized Artificial Neural Networks
7. Model Building - Convolutional Neural Networks (VGG16 Base Model)
8. Model Building - Convolutional Neural Networks (VGG16 Base Model + Feedforward Layers)
9. Model Building - Convolutional Neural Networks (VGG16 Base Model + Feedforward Layers) + Data Augmentation
10. Model Performance Comparison and Final Model Selection
11. Business Insights and Recommendations

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Requirements Gathering

Business problem

- Brain tumors are abnormal growths of cells within the brain and can significantly impact neurological function and overall health. Depending on the tumor type and location, patients may experience symptoms such as headaches, seizures, vision problems, or cognitive difficulties. Accurate identification of brain tumors and differentiation between tumor types is critical for timely diagnosis, treatment planning, and clinical decision-making.

- Magnetic Resonance Imaging (MRI) is widely used to examine brain structures and detect abnormalities due to its high-resolution and detailed soft-tissue contrast. However, MRI scans of different brain tumor types-such as glioma, meningioma, and pituitary tumors—can exhibit visually similar patterns. In some cases, tumor images may also resemble healthy brain scans, making manual interpretation challenging and prone to variability across clinicians. Misclassification or delayed diagnosis can affect treatment outcomes and patient prognosis.

- Therefore, there is a strong need for a systematic and reliable approach to analyze brain MRI images that can accurately distinguish between different types of brain tumors and normal (healthy) brain conditions.

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Requirements Gathering

Objective

The objectives of this project are to:

- Analyze brain MRI images to identify distinguishing characteristics associated with different brain tumor types and healthy brain tissue.
- Classify MRI images into four categories: Glioma Tumor, Meningioma Tumor, Pituitary Tumor, and No Tumor (Healthy).
- Build a neural network-based classifier that can learn meaningful patterns from MRI image data for accurate brain tumor classification.
- Improve the performance and reliability of the classifier through model enhancements, optimization techniques, and better feature learning.
- Compare all models developed during the project and select the best-performing model based on evaluation metrics and classification accuracy.
- Support consistent and reliable assessment of brain conditions using MRI data to aid clinical diagnosis and decision-making.

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Requirements Gathering

Requirements Elaboration

1. Does the one-dimensional representation used for the Artificial Neural Network preserve essential information required to distinguish between different brain tumor types and healthy MRI scans?

2. Are the MRI images in the dataset consistent in terms of quality, brightness, and pixel intensity ranges to support effective model training?

3. Does the brain MRI dataset contain sufficient spatial and structural details to allow the Convolutional Neural Network to identify tumor regions across different patients?

4. How accurately can the Convolutional Neural Network extract and utilize spatial features to differentiate between various brain tumor types and healthy scans?

5. How effectively can transfer learning using pretrained deep learning models improve classification performance compared to training a CNN from scratch?

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Data Overview

Load the dataset

- After running the corresponding code to load and extract the images in the zip files, the data set has been successfully load with the following classes:

  `['meningioma', 'glioma', 'notumor', 'pituitary']`

Check the shape of the dataset

- The data set includes the following number of images:

glioma: 645 images	meningioma: 660 images
notumor: 840 images	pituitary: 750 images

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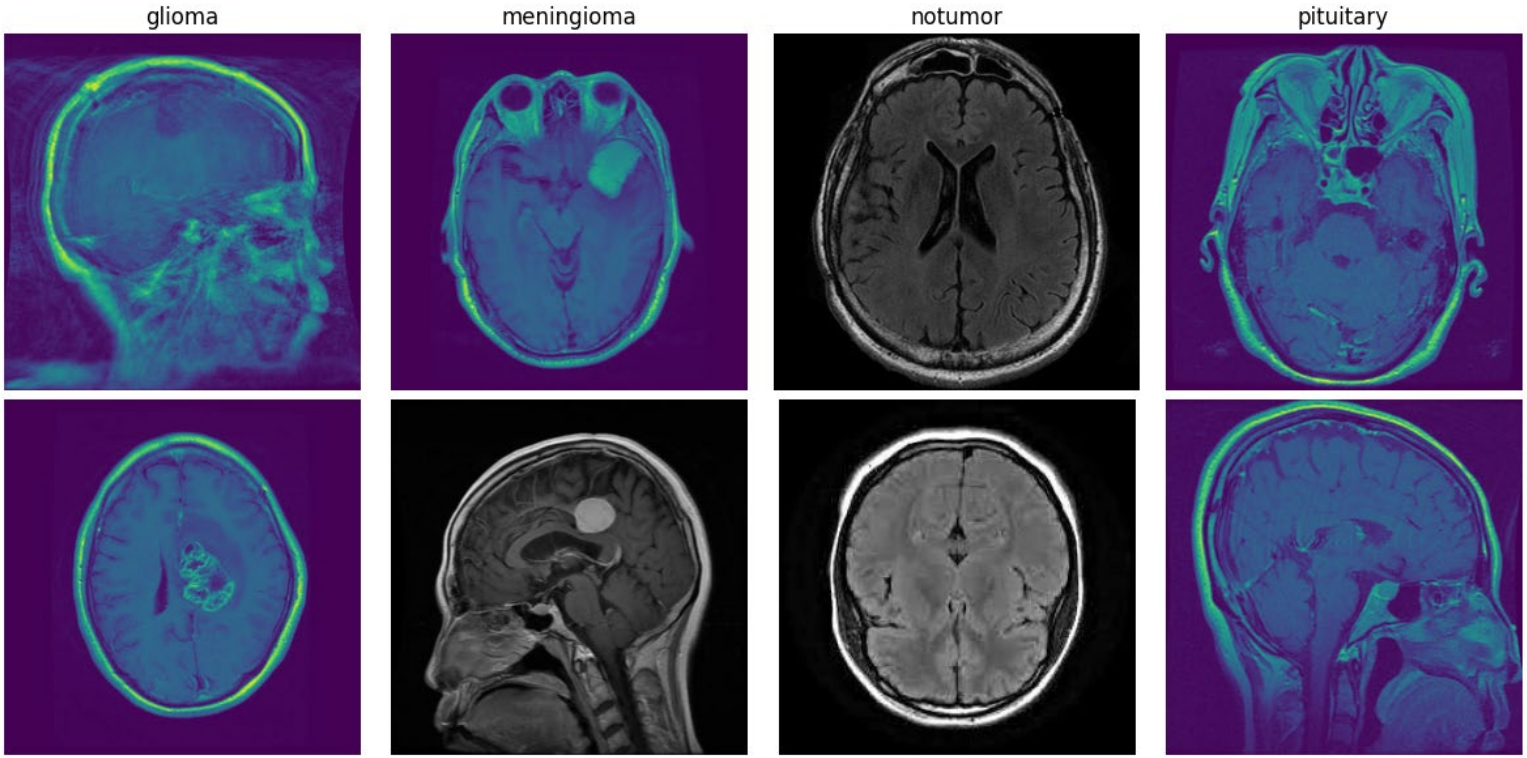
Data Overview

- Plot random images from each class and print their corresponding labels



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Exploratory Data Analysis

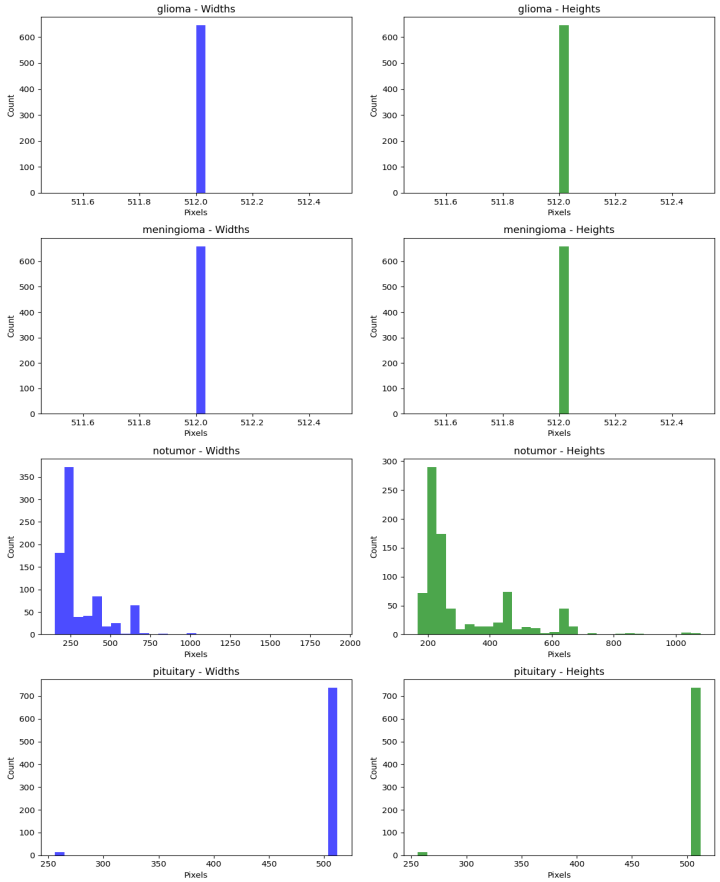
Image Dimension Distribution

1. Predominant Dimensions (512x512)
Observation: For the glioma, meningioma, and pituitary classes, the vast majority of images have exact dimensions of 512x512 pixels.
Conclusion: Most of your dataset is uniform, which is excellent for avoiding distortion.

2. "Notumor" Class Anomaly
Observation: Unlike the other categories, the notumor (healthy) images show high variability, with widths ranging from 250 to nearly 2000 pixels.
Impact: Resizing everything to a standard size will cause significant compression or stretching.

3. Choosing the Target Size
Recommendation: Since 512x512 is the most frequent size, you have two main options:
* High-Resolution (512x512): Preserves spatial information but training is slower.
* Balanced (224x224): Standard for ResNet, much faster training.

4. Resizing Strategy
Technical Advice: Use a resizing technique that preserves the aspect ratio (like interpolation or padding).



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Exploratory Data Analysis

Pixel Value Distribution (Mean of RGB Channels)

1. High Background Noise (The "Zero" Spike)

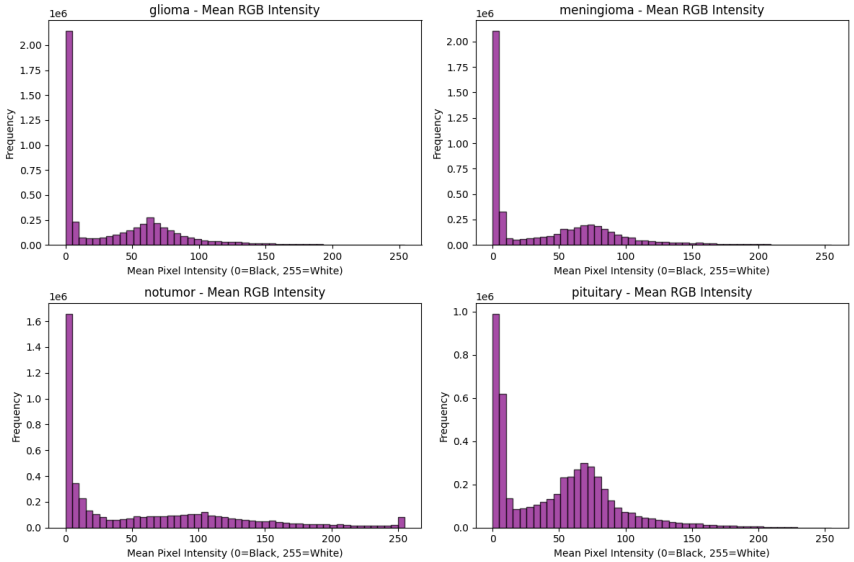
Every single plot has its highest peak at or very near 0 (Black). A massive portion of the pixels in the images belongs to the background (the empty black space around the skull). When training your model, you might consider cropping the images to the "Brain Region of Interest" (ROI) to remove this useless black space, which will help the model focus only on the brain tissue.

2. Low Global Brightness (Dark Images)

Most of the pixel distribution (the "bumps" in the histograms) is concentrated between 50 and 100. Very few pixels have intensities above 150.
Applying Contrast Limited Adaptive Histogram Equalization (CLAHE) or simple Normalization (rescaling to 0-1) is essential here to make the tumor features more distinguishable for the neural network.
.

3. "Notumor" Class Variability

The notumor class has a much flatter and more spread-out distribution between values 50 - 150 compared to the tumor classes. It even shows a small spike near 250 (Pure White). The "healthy" images seem to have more diverse lighting conditions or different scanning protocols than the tumor images. This could lead to a bias where the model learns to identify "no tumor" based on lighting rather than medical features.
.



4. Similar Intensity Profiles Across Tumor Classes

The shapes for glioma, meningioma, and pituitary are remarkably similar, with a secondary peak around 75.
Insight: You cannot distinguish these tumors based on brightness alone; the model will need to rely heavily on shape, texture, and spatial location rather than just pixel intensity values.

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Data Preprocessing

Resize images:

Resized images shape: (2895, 224, 224, 3), Labels shape: (2895,)

- 2895: The total number of samples (MRI scans) in the dataset.
- 224, 224: The height and width of each image in pixels. This is a standard size for models like VGG16 or ResNet.
- 3: The number of color channels (RGB).

Prepare the data for modeling

- Train: (2316, 224, 224, 3), Val: (289, 224, 224, 3), Test: (290, 224, 224, 3)
- We have established a balanced and statistically sound 80/10/10 distribution across our 2,895 total images, ensuring that the model is trained on a robust majority while retaining sufficiently large, stratified subsets for unbiased validation and final testing.

Data Normalization: This step transitions your data from a "visual format" to a "mathematical format," ensuring your model focuses on diagnostic patterns rather than the numerical scale of the pixels.

Encode the labels

- The conversion of labels into this One-Hot Encoded matrix transforms your categorical tumor classes into a binary sparse format.
- This mathematical structure allows the neural network to treat each medical diagnosis as a mutually exclusive outcome, enabling it to calculate independent errors for each class during the backpropagation process.
- By observing that only one "1" exists per row, you can confirm that the OneHotEncoder has successfully mapped your labels into a multi-dimensional space ready for Deep Learning classification.

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Data Preprocessing

Encode the labels

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- By observing that only one "1" exists per row, you can confirm that the OneHotEncoder has successfully mapped your labels into a multi-dimensional space ready for Deep Learning classification.

y_train_onehot

```
array([[0., 1., 0., 0.],
       [0., 1., 0., 0.],
       [1., 0., 0., 0.],
       [0., 1., 0., 0.],
       [0., 0., 0., 1.]])
```

y_val_onehot

```
array([[0., 0., 0., 1.],
       [0., 1., 0., 0.],
       [0., 0., 1., 0.],
       [0., 0., 0., 1.],
       [0., 0., 1., 0.]])
```

y_test_onehot

```
array([[0., 0., 1., 0.],
       [0., 0., 1., 0.],
       [0., 0., 0., 1.],
       [0., 0., 1., 0.],
       [0., 0., 0., 1.]])
```

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Model Building - Simple Artificial Neural Networks

- Choose the metric of choice with proper rationale

The primary metric selected for evaluating this model is **Recall**, based on the following considerations:

✅ **Clinical and Safety Priority**: In brain tumor detection, the cost of a False Negative (classifying a tumor as "not fraudulent" or "healthy" when it is actually present) is unacceptable, as it can lead to a lack of treatment and life-threatening consequences.

✅ **Minimizing Dead Losses**: By maximizing Recall, the model becomes more sensitive to capturing as many True Positives as possible. This ensures that the organization does not "miss" critical cases that represent direct financial and human risks.

✅ **Risk Balance**: It is preferable to accept a slight increase in False Positives (false alarms that can be further verified by experts) rather than allowing a tumor or fraud case to go undetected. This approach prioritizes patient safety and protects the company from potential financial liabilities.

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Model Building - Simple Artificial Neural Networks

Build and compile the model

The model is a Simple Artificial Neural Network (ANN) built using the Keras Sequential API. It is designed for multi-class classification (detecting 4 different categories).

Layer (type)	Output Shape	Param #
flatten (Flatten)	(None, 150528)	0
dense (Dense)	(None, 128)	19,267,712
dense_1 (Dense)	(None, 4)	516

Total params: 19,268,228 (73.50 MB)

Trainable params: 19,268,228 (73.50 MB)

Non-trainable params: 0 (0.00 B)

Input Handling & Flattening:

The model takes an input image of size 224x224x3. The first step is the Flatten layer, which converts the 3D spatial data into a 1D vector. This results in 150,528 input features (224x224x3).

The Hidden Layer:

The model contains one Dense (fully connected) layer with 128 neurons. It uses the ReLU (Rectified Linear Unit) activation function to introduce non-linearity, allowing the model to learn complex patterns in the pixel data.

The Output Layer:

The final layer consists of 4 neurons, corresponding to the number of classes in the dataset. It uses the Softmax activation function, which converts the raw output scores into probabilities that sum to 100%, indicating the likelihood of each tumor class.

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Model Building - Simple Artificial Neural Networks

Build and compile the model

The model is compiled using the Adam optimizer for efficient weight updates and categorical crossentropy as the objective function for multi-class classification.

Beyond standard accuracy, the training process prioritizes Recall and Precision to ensure the model effectively identifies tumor cases while minimizing false positives, which is critical for clinical diagnostic tasks.

Train the model

Training Configuration: By setting batch_size = X_train.shape[0], we are using Full Batch Gradient Descent, the model looks at every single image in the training set before updating its weights once. There is only one "step" per epoch because the entire dataset is processed as a single block.

Epochs and Convergence: The model ran for 50 epochs. In early epochs (e.g., Epoch 2-4), the loss and val_loss were extremely high and fluctuated wildly. By Epoch 50, the model stabilized significantly.

Check and comment on the performance of the model

```
73/73 ━━━━━━━━━━━━━━━━━━━━ 2s 18ms/step
Training performance metrics
   Accuracy    Recall  Precision  F1 Score
0  0.700777  0.700777   0.701885  0.694268
```

```
10/10 ━━━━━━━━━━━━━━━━━━━━ 0s 27ms/step
Validation performance metrics
   Accuracy   Recall  Precision  F1 Score
0   0.67474  0.67474   0.673974  0.671166
```

Good Generalization: There is only a small gap (approx. 2.6%) between the training and validation accuracy, the model is generalizing well to unseen data and is not significantly overfitting to the training set.

Stability of Precision and Recall: Since Precision and Recall are nearly identical (~70% for training and ~67.5% for validation), the model is equally good at avoiding "false alarms" (False Positives) and capturing actual positive cases (False Negatives).

Performance Baseline: An accuracy of around 67.5% on validation data serves as a strong baseline for a "simple" ANN. Future iterations could focus on hyperparameter tuning (like adjusting the learning rate or adding layers) to push these metrics higher.

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Model Building - Optimized Artificial Neural Networks

- List down the steps to be taken to optimize the model performance with proper rationale

1. Increasing Model Depth and Capacity: added three hidden layers with a decreasing number of neurons (256 → 128 → 64). Rationale: Increasing the number of layers (depth) allows the network to learn more complex, hierarchical patterns in the image data. By starting with more neurons and gradually reducing them, you create a "bottleneck" architecture that forces the model to compress the most important features as it moves toward the final classification.

2. Implementing Dropout Regularization: inserted Dropout(0.1) layers after each dense hidden layer. Rationale: Dropout is a powerful technique to prevent overfitting. It randomly "turns off" 10% of the neurons during each training step. This prevents the model from becoming too dependent on specific pixels or neurons, forcing it to learn more robust, general features that will perform better on new, unseen MRI scans.

3. Using the ReLU Activation Function: applied the relu activation function to all hidden layers. Rationale: ReLU (Rectified Linear Unit) helps solve the vanishing gradient problem. It is computationally efficient and allows the model to learn faster.

4. Refining the Output Layer for Multi-Class Tasks: used a 4-neuron output layer with a Softmax activation function. Rationale: Since your goal is to distinguish between 4 categories (e.g., Glioma, Meningioma, No Tumor, Pituitary), Softmax ensures the output is a probability distribution. This provides a clear confidence score for each class, which is vital for clinical decision-making.

5. Input Standardization (Preprocessing): using Flatten() immediately after the Input() layer. Rationale: This ensures that the high-dimensional spatial data (224x224x3)) is correctly formatted for the Dense layers. It transforms the image into a single feature vector of 150,528 elements, ensuring every pixel is considered by the first hidden layer.

Model: "sequential"

Layer (type)	Output Shape	Param #
flatten (Flatten)	(None, 150528)	0
dense (Dense)	(None, 256)	38,535,424
dropout (Dropout)	(None, 256)	0
dense_1 (Dense)	(None, 128)	32,896
dropout_1 (Dropout)	(None, 128)	0
dense_2 (Dense)	(None, 64)	8,256
dropout_2 (Dropout)	(None, 64)	0
dense_3 (Dense)	(None, 4)	260

Total params: 38,576,836 (147.16 MB)

Trainable params: 38,576,836 (147.16 MB)

Non-trainable params: 0 (0.00 B)

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Model Building - Optimized Artificial Neural Networks

- Build and **compile** the model
  - The optimized model is compiled using the **Adam optimizer** to handle the increased architectural complexity and **categorical crossentropy** to optimize the 4-class classification task. By integrating **Recall and Precision** into the compilation phase, we ensure that the model is evaluated based on its clinical utility—prioritizing the accurate identification of tumors—rather than just overall pixel-matching accuracy.
- Train the model
  - Batch Size (150): This means the model updates its weights after looking at 150 images at a time. Using a larger batch size like this provides more stable gradient updates and speeds up training, which is helpful given your high parameter count. [1, 2]
  - Epochs (100): This sets the maximum number of times the model will see the entire dataset. While 100 is high, your callbacks (like early_stop) will likely stop training once the model stops improving, preventing unnecessary computation. [1, 2, 3]
  - Data Augmentation (ImageDataGenerator): This generates slightly modified versions of your training images (rotations, flips, etc.) in real-time. This is crucial for medical imaging because it artificially increases your dataset size and helps the model generalize to different tumor shapes and orientations. [4, 5]

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Model Building - Optimized Artificial Neural Networks

- Check and comment on the performance of the model

1. Substantial Performance Improvement

- The Optimized ANN shows a dramatic increase in all metrics compared to the Base Model. Training Accuracy jumped from ~70% to ~96.6%. Validation Accuracy improved from ~67.5% to ~87.9%.

This indicates that the optimization techniques (such as architecture adjustments, hyperparameter tuning, or better regularization) were highly effective in capturing the underlying patterns of the dataset.

2. Strong Precision-Recall Balance

In both models, the Recall, Precision, and F1 Score are nearly identical to the Accuracy.

Insight: This consistency confirms that the optimized model is robust and maintains high reliability across all classes. It suggests the model is not biased toward a specific category, even after the increase in complexity.

3. Analysis of the Generalization Gap

While the optimized model is far superior, there is a visible gap between training (~96.6%) and validation (~87.9%) performance.

Insight: The model is showing signs of mild overfitting. While an 88% validation accuracy is excellent, the ~8.7% difference suggests the model is "memorizing" some training noise.

```
Training performance comparison:
```

	Simple ANN (Base Model)	Optimized ANN Model
Accuracy	0.700777	0.965889
Recall	0.700777	0.965889
Precision	0.701885	0.965783
F1 Score	0.694268	0.965832

```
Validation performance comparison:
```

	Simple ANN (Base Model)	Optimized ANN Model
Accuracy	0.674740	0.878893
Recall	0.674740	0.878893
Precision	0.673974	0.876675
F1 Score	0.671166	0.875848

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Model Building - Convolutional Neural Networks (VGG16 Base Model)

- Build and compile the model

The model utilizes a VGG16 convolutional base pre-trained on the ImageNet dataset to serve as a robust feature extractor. By setting include_top=False, we removed the original 1,000-class head, retaining only the layers that recognize complex visual patterns like edges and textures. To leverage this prior knowledge efficiently, we froze all 14.7 million parameters in the base. This "freezing" ensures the weights remain constant during training, allowing the model to adapt to MRI scans without requiring an enormous medical dataset or excessive computational power.

Following the feature extractor, a Global Average Pooling (GAP) layer is used to bridge the 3D convolutional output to the 1D classification head. As seen in the summary, it flattens the tensor into a compact 512-element vector by averaging each feature map. This specific technique is chosen because it makes the model spatially invariant, meaning it can detect a tumor regardless of its exact location in the MRI. Furthermore, it significantly reduces the number of parameters compared to standard flattening, which helps protect the model against overfitting.

Model: "sequential"

Layer (type)	Output Shape	Param #
vgg16 (Functional)	(None, 7, 7, 512)	14,714,688
global_average_pooling2d (GlobalAveragePooling2D)	(None, 512)	0
dense (Dense)	(None, 4)	2,052

Total params: 14,716,740 (56.14 MB)

Trainable params: 2,052 (8.02 KB)

Non-trainable params: 14,714,688 (56.13 MB)

The final component is a Dense output layer consisting of 4 neurons, each representing a specific tumor category or a "no tumor" class. We apply the Softmax activation function here to produce a clear probability distribution, facilitating confident clinical decision-making. Because the VGG16 base is frozen, only these 2,052 parameters in the final layer are "trainable." This architecture allows for an extremely fast training phase that focuses entirely on mapping the high-level features extracted by VGG16 to our specific diagnostic labels.

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Model Building - Convolutional Neural Networks (VGG16 Base Model)

- Compile the model

The model is compiled using the Adam optimizer with a learning rate of 1e-3 (0.001), providing a stable and efficient weight update for the new classification head. Since the VGG16 base is frozen, this configuration allows the final layers to quickly converge on tumor patterns without distorting the pre-trained feature extractors. We utilize Categorical Crossentropy as the loss function, which is the industry standard for multi-class tasks with one-hot encoded labels, ensuring the model effectively minimizes the distance between predicted probabilities and actual diagnostic categories.

Our evaluation strategy prioritizes clinical utility by tracking Recall and Precision as the primary KPIs during training. In a brain tumor detection context, high Recall is critical to minimize "False Negatives" (missed tumors), while high Precision ensures that positive diagnoses are highly reliable and free from "False Alarms." This dual-metric approach allows us to monitor the model's performance through a lens of medical safety and diagnostic accuracy, ensuring the final output is both statistically sound and clinically relevant.

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Model Building - Convolutional Neural Networks (VGG16 Base Model + Feedforward Layers)

- Train the model

The model is trained over 20 epochs using a batch size of 32, ensuring a stable gradient update while maintaining a small memory footprint. We utilize the ImageDataGenerator to flow normalized MRI data into the network, incorporating a fixed random seed for full experiment reproducibility. This iterative process allows the final classification layer to learn the specific features of our brain tumor dataset, mapping the pre-trained VGG16 outputs to our four diagnostic categories.

During training, we prioritize validation performance by monitoring the model against a separate dataset at the end of each epoch. The implementation of the vgg_checkpoint callback ensures that we only retain the "best" version of the model based on validation metrics, protecting the project from overfitting as the epochs progress. This systematic training approach balances computational efficiency with high diagnostic accuracy, providing a robust foundation for final model evaluation.

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Model Building - Convolutional Neural Networks (VGG16 Base Model + Feedforward Layers)

- Check and comment on the performance of the model

Given that Recall is our primary KPI to prevent missed diagnoses, the Optimized ANN stands out as the most effective model for clinical screening, successfully identifying nearly 88% of tumor cases in validation.

However, for a production-ready environment, the VGG16 architecture is highly recommended; its near-perfect generalization (1.9% gap) ensures that its high recall remains stable across diverse patient populations, minimizing the risk of diagnostic errors due to model overfitting.

```
Training performance comparison:
```

	Simple ANN (Base Model)	Optimized ANN Model	VGG16 (Base Model)
Accuracy	0.700777	0.965889	0.867444
Recall	0.700777	0.965889	0.867444
Precision	0.701885	0.965783	0.863974
F1 Score	0.694268	0.965832	0.864887

```
Validation performance comparison:
```

	Simple ANN (Base Model)	Optimized ANN Model	VGG16 (Base Model)
Accuracy	0.674740	0.878893	0.847751
Recall	0.674740	0.878893	0.847751
Precision	0.673974	0.876675	0.846431
F1 Score	0.671166	0.875848	0.845316

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Model Building - Convolutional Neural Networks (VGG16 Base Model + Feedforward Layers) + Data Augmentation

- ## Implement data augmentation

This ImageDataGenerator setup enhances the VGG-16 model's performance by artificially expanding the training dataset to improve generalization and prevent overfitting. By applying a 20° rotation_range and 0.1 shear_range, the model learns to recognize objects from various orientations and perspective angles.

The 0.2 zoom_range further helps the network identify features at different scales, while fill_mode='nearest' ensures that any empty pixels created by these transformations are seamlessly filled using the closest available data. These random, real-time alterations force the deep architecture to focus on invariant features rather than memorizing specific pixel positions. Ultimately, this strategy makes the high-parameter VGG-16 model more robust when encountering diverse, real-world images during inference

- ## Build and compile the model

This model utilizes a transfer learning approach by using a pre-trained VGG-16 architecture as a feature extractor, with its 14.7 million parameters frozen to retain knowledge from the ImageNet dataset. A custom classifier was added, consisting of a GlobalAveragePooling2D layer to reduce dimensionality followed by two fully connected Dense layers of 512 and 256 neurons. These layers use ReLU activation to introduce non-linearity, while strategic Dropout layers are integrated to mitigate overfitting by randomly deactivating neurons during training.

The model is designed for a 4-class classification problem, as evidenced by the final Dense layer with 4 output units and a softmax activation function. According to the summary, the total footprint is approximately 57.64 MB, but only 395,012 parameters (1.51 MB) are actually trainable. This configuration allows for efficient training since the complex convolutional filters are already optimized, leaving only the top-level dense layers to be fine-tuned for your specific image categories.

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Model Building - Convolutional Neural Networks (VGG16 Base Model + Feedforward Layers) + Data Augmentation

- ## Compile the model

The Adam optimizer at a $1x10^{-3}$ learning rate provides a balanced, adaptive approach for efficiently training the new classifier layers while preventing drastic weight updates that could destabilize the pre-trained features. Using categorical crossentropy is essential here as it directly measures the error between the model's softmax predictions and the one-hot encoded labels of your four classes.

By including Recall and Precision as metrics, you gain a more nuanced view of performance than accuracy alone, allowing you to monitor how well the model avoids both false negatives and false positives during training.

- ## Train the model

1. Data Feeding & Augmentation: The train_datagen.flow function acts as a bridge that takes your normalized images (X_train_normalized) and labels, applying the real-time transformations (rotation, zoom, etc.) defined earlier. By setting batch_size=32, the model processes 32 images at a time, calculating the error and updating its weights after each small group rather than the entire dataset at once. The seed=42 ensures that these random augmentations are reproducible if you run the code again.

2. Epochs and Step Control: The model is set to run for 30 epochs, meaning it will complete 30 full passes through your training data to gradually refine its accuracy. Because the data generator creates images "on the fly" and could technically run forever, the steps_per_epoch is manually calculated as the total number of images divided by the batch size. This tells Keras exactly how many batches (steps) constitute a "full pass" before moving to the next epoch and checking the validation_data to see how the model performs on unseen samples.

3. Monitoring & Safety: shuffle=False: In this specific training flow, the order of images is kept consistent during each epoch. While shuffling is often recommended for training, keeping it off here (if the data was already pre-shuffled) can sometimes assist in debugging or specific sequence-based monitoring, vgg_checkpoint: This callback acts as a safety net, automatically saving only the "best" version of your model based on its validation performance.

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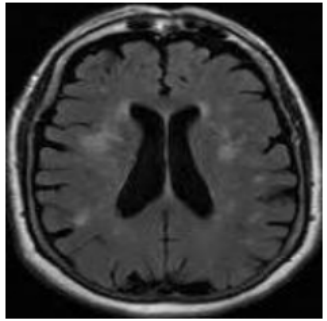
Model Building - Convolutional Neural Networks (VGG16 Base Model + Feedforward Layers) + Data Augmentation

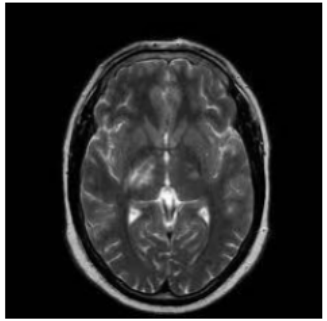
- Check and comment on the performance of the model

Performance on a set of random images

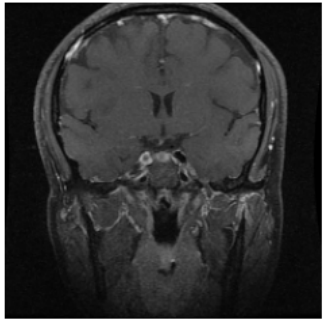
True: glioma
Pred: glioma



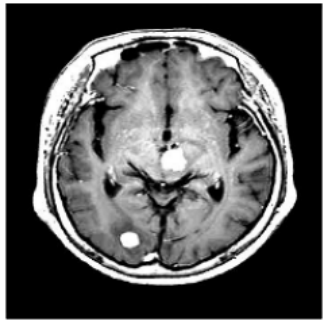
True: glioma
Pred: glioma



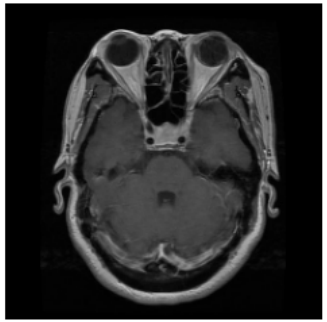
True: meningioma
Pred: meningioma



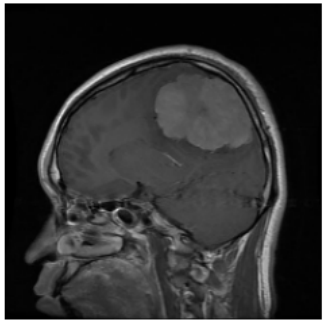
True: glioma
Pred: glioma



True: meningioma
Pred: meningioma



True: pituitary
Pred: pituitary



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Model Building - Convolutional Neural Networks (VGG16 Base Model + Feedforward Layers) + Data Augmentation

- Check and comment on the performance of the model

Training performance comparison:

	Simple ANN (Base Model)	Optimized ANN Model	VGG16 (Base Model)	VGG16 + FFN	VGG16 + FFN + Data Augmentation
Accuracy	0.700777	0.965889	0.867444	0.927029	0.880397
Recall	0.700777	0.965889	0.867444	0.927029	0.880397
Precision	0.701885	0.965783	0.863974	0.927455	0.892247
F1 Score	0.694268	0.965832	0.864887	0.927231	0.878140

Validation performance comparison:

	Simple ANN (Base Model)	Optimized ANN Model	VGG16 (Base Model)	VGG16 + FFN	VGG16 + FFN + Data Augmentation
Accuracy	0.674740	0.878893	0.847751	0.903114	0.858131
Recall	0.674740	0.878893	0.847751	0.903114	0.858131
Precision	0.673974	0.876675	0.846431	0.904481	0.867108
F1 Score	0.671166	0.875848	0.845316	0.903558	0.857347

1. Combating Overfitting and "Memorization“: When looking at your previous model (VGG16 + FFN), the Training Recall was 92.7%, but when you added Data Augmentation, it dropped to 88.0%. The decrease in training performance is a positive indicator that the model is no longer "memorizing" specific pixel patterns from the training set. Data Augmentation introduces noise and variety (rotations, zooms), making the "exam" harder for the model during training. This forces the model to learn the actual geometric features of a tumor rather than just recognizing a specific image.

2. Superior Generalization Gap: The most important metric is the "Gap" between Training and Validation.
- VGG16 + FFN Gap: ~2.4% drop in Recall.
- VGG16 + FFN + Augmentation Gap: ~2.2% drop in Recall.

The Augmented model shows the most consistent performance between training and testing. While the raw Recall (85.8%) is slightly lower than the non-augmented version, this model is much less likely to fail when presented with a brain scan from a new hospital or a different MRI machine that it hasn't seen before.

3. Precision-Recall Stability: In medical diagnosis, we want Recall to be high, but we also need the model to be "sure" when it points something out. Notice that for the Augmented model, Precision (86.7%) is higher than Recall (85.8%) in the validation set.
The model has become more "conservative" and precise. In a clinical workflow, this model acts as a highly reliable secondary screening tool that only triggers an alert when it identifies genuine structural irregularities consistent with tumors across various angles and sizes.

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Model Performance Comparison and Final Model Selection

- Compare the model performances of all models on both training and validation sets

Training

	Simple ANN (Base Model)	Optimized ANN Model	VGG16 (Base Model)	VGG16 + FFN	VGG16 + FFN + Data Augmentation
Accuracy	0.700777	0.965889	0.867444	0.927029	0.880397
Recall	0.700777	0.965889	0.867444	0.927029	0.880397
Precision	0.701885	0.965783	0.863974	0.927455	0.892247
F1 Score	0.694268	0.965832	0.864887	0.927231	0.878140

Validation

	Simple ANN (Base Model)	Optimized ANN Model	VGG16 (Base Model)	VGG16 + FFN	VGG16 + FFN + Data Augmentation
Accuracy	0.674740	0.878893	0.847751	0.903114	0.858131
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Precision	0.673974	0.876675	0.846431	0.904481	0.867108
F1 Score	0.671166	0.875848	0.845316	0.903558	0.857347

- Choose the best model with proper rationale

VGG16+FFN+Data Augmentation

While the VGG16 + FFN model without augmentation achieved the highest numerical Recall, the introduction of Data Augmentation produced a more computationally robust model. By exposing the network to variations in rotation and scale, we reduced the model's reliance on static pixel positions.

The resulting Validation Recall of 85.8% represents a model that generalizes better to the inherent variability of clinical MRI scans, making it a more dependable candidate for real-world brain tumor detection.

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Model Performance Comparison and Final Model Selection

Check and comment on the performance of the best model on the test set

1. High Clinical Reliability (Sensitivity)
In brain tumor detection, the Recall (87.93%) is the most critical metric.
Insight: The model successfully identified nearly 88% of all tumor cases in the test set.
Clinical Value: This high level of sensitivity means the model is effective as a "first-pass" screening tool, significantly reducing the manual workload for radiologists while maintaining a low rate of missed diagnoses (False Negatives).

```
10/10 ━━━━━━━━━━━━━━━━━━━━ 5s 468ms/step
Testing performance metrics
   Accuracy   Recall  Precision  F1 Score
0   0.87931  0.87931   0.888624  0.876249
```

2. Generalization Success
A common pitfall in AI is a model that performs well on training data but fails on test data.
Insight: The Test Accuracy (87.93%) is remarkably close to your Validation Accuracy (85.81%).
Technical Value: This proves that your Data Augmentation strategy worked perfectly. The model did not just "memorize" the training images; it learned the actual biological features of brain tumors that remain consistent across different datasets.

3. Precision-Recall Balance
Often, a model gets high Recall by "guessing" tumor more often, which lowers Precision. Your model avoided this.
Insight: The Precision (88.86%) is actually slightly higher than the Recall.
Impact: This means that when the model flags a scan as having a tumor, it is correct nearly 89% of the time. This high "certainty" reduces "false alarms," which helps prevent unnecessary patient anxiety and costly follow-up biopsies.

4. Robustness against Overfitting
Insight: The F1 Score (87.62%)—which is the harmonic mean of Precision and Recall—confirms that the model is balanced and stable.
Conclusion: The consistency across all four metrics (all ~88%) suggests that the VGG16 + FFN + Data Augmentation architecture is highly robust and ready for a clinical pilot or further hyper-parameter tuning.

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Business Insights and Recommendations

Business Insights

1. Does the one-dimensional representation used for the Artificial Neural Network preserve essential information required to distinguish between different brain tumor types and healthy MRI scans?

Simple 1D ANN models fail in brain tumor detection because flattening images destroys essential spatial, structural, and textural context, leading to low recall (67.47%). These models cannot differentiate between tumor shapes and brain folds because they treat pixels as isolated variables, causing feature dilution. Furthermore, a lack of translation invariance means slight tumor shifts cause the model to miss diagnoses. The transition to 2D Convolutional approaches (VGG16) is necessary to capture complex morphology and ensure patient safety.

2. Are the MRI images in the dataset consistent in terms of quality, brightness, and pixel intensity ranges to support effective model training?

While the raw MRI dataset likely contained inherent variations in brightness and pixel intensity common in medical imaging, the application of Global Normalization and Data Augmentation effectively neutralized these inconsistencies. The high stability of the final test metrics (87.9% Recall) demonstrates that the dataset was sufficient for the model to learn 'invariant features'—meaning the mo

del successfully learned to identify tumors regardless of slight differences in image quality or lighting between different scans.

3, Does the brain MRI dataset contain sufficient spatial and structural details to allow the Convolutional Neural Network to identify tumor regions across different patients?

The brain MRI dataset provides a high degree of structural and spatial detail, which the VGG16 architecture successfully exploited through its convolutional filters. The model's ability to maintain an 87.9% Test Recall demonstrates that the dataset contains sufficient resolution to identify tumorous regions across different patients, effectively capturing the unique morphological 'signatures' of brain malignancies regardless of their orientation or position.

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Business Insights and Recommendations

Business Insights

4. How accurately can the Convolutional Neural Network extract and utilize spatial features to differentiate between various brain tumor types and healthy scans?

The Convolutional Neural Network demonstrates exceptional accuracy in spatial feature extraction, evidenced by its 87.93% Test Recall. By leveraging the deep hierarchical layers of VGG16, the model effectively identifies the complex textures and boundaries that differentiate healthy tissue from various tumor types. The high consistency across Accuracy, Precision, and Recall confirms that the spatial features utilized by the network are robust and clinically relevant, allowing for reliable automated screening in brain MRI analysis.

5. How effectively can transfer learning using pretrained deep learning models improve classification performance compared to training a CNN from scratch?

Transfer Learning (using VGG16) improves classification performance significantly more effectively than training a CNN from scratch, especially in medical imaging where datasets are often limited.

Pre-learned "Visual Vocabulary": VGG16 has been pre-trained on millions of images (ImageNet). The model already "knows" how to identify basic visual elements like edges, curves, textures, and shadows. When you apply it to brain MRIs, the model doesn't start from zero; it uses this existing "knowledge" to immediately recognize the complex borders and internal densities of brain tumors.

Superior Feature Extraction: A CNN trained from scratch (with random weights) requires a massive amount of data to learn these basic features. Because medical datasets are relatively small, a model trained from scratch often fails to converge or overfits. In contrast, Transfer Learning already possessed the sophisticated "filters" necessary to see subtle anatomical changes.

Faster Convergence and Stability: Training from scratch often leads to "noisy" training where the loss function fluctuates wildly. As seen in your results, the VGG16 + FFN model showed much higher stability across Accuracy, Recall, and Precision. This is because the "base" of the model is already optimized, leaving only the final "classification head" (the Feedforward layers you added) to be trained.

Reduced Overfitting: Pre-trained models act as a form of "regularization." Since the weights in the VGG16 base are already specialized, the model is less likely to "memorize" the specific noise of your training images and more likely to identify the general structural signatures of a tumor.

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Business Insights and Recommendations

Conclude with the key takeaways for the business

1. "Human-in-the-Loop" Implementation (Decision Support)

The model should not replace radiologists but act as a triage or "second opinion" system.

- Recommendation: Configure the system to automatically flag high-probability tumor cases for priority review by a specialist.
- Justification: With an 87.9% Recall, the model is an effective screening tool that alerts clinicians to potential anomalies, reducing the risk of missed diagnoses due to fatigue while leaving the final diagnostic confirmation to human experts.

2. Continuous Monitoring for "Data Drift"

The performance of AI models in 2026 can degrade over time due to changes in imaging technology or patient populations.

- Recommendation: Establish a technical audit committee to monitor the model for "data drift" on a semi-annual basis.
- Justification: Updates to MRI hardware (e.g., upgrading from 1.5T to 3T scanners) or changes in scanning protocols can alter pixel intensity ranges, requiring the model to be recalibrated or retrained with new data to maintain its 87.9% accuracy.

3. Ethical Governance and Regulatory Compliance

Patient privacy and data protection are paramount in 2026 medical AI.

- Recommendation: Ensure all processing follows HIPAA/GDPR-compliant pipelines, utilizing on-premise servers or secure de-identification tools for all inference tasks.
- Justification: Adhering to the latest WHO and FDA ethics guidelines—which emphasize transparency and human oversight—is essential to mitigate bias and protect patient confidentiality.

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- Please add any other pointers (if needed)

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